Gulaba Ram

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Professional Summary

Information Technology undergraduate at NIT Srinagar with hands-on experience building full-stack and back-end systems with GenAI/LLM integration in JavaScript and Python, and a strong DSA foundation in C++; seeking software engineering roles.

Education

- **National Institute of Technology, Srinagar** **2023 - 2027**
  B.Tech in Information Technology CGPA: 8.2/10
- **Rajasthan Board of Secondary Education 12th** **2021 - 2022**
  Senior School Examination Percentage: 94.20%

Projects

AI Database Agent (Text-to-SQL) Source Code June 2026
FastAPI, LangGraph, PostgreSQL, SQLAlchemy, React, Docker, AWS

- Built a full-stack text-to-SQL agent that compiles natural-language questions into safe, read-only queries through a multi-node LangGraph pipeline.
- Enforced query governance with an sqlglot AST validator – SELECT-only, auto-injected row limits and statement timeouts – blocking unsafe SQL before it reaches the database; secured access with JWT + RBAC and Fernet-encrypted credentials at rest.
- Containerized FastAPI, React, PostgreSQL, Redis, and Nginx with Docker Compose and deployed to AWS EC2 behind a reverse-proxy gateway over an async SQLAlchemy backend with Alembic migrations.

Accident Risk Prediction August 2026
Python, Pandas, Scikit-learn, XGBoost

- Developed a machine learning system to predict road accident risk from speed, weather, traffic, road condition, time, and historical accident data, using preprocessing and engineered features such as rush-hour indicators and weather severity.
- Trained and compared Logistic Regression, Random Forest, and XGBoost classifiers, and built a dashboard for real-time Low / Medium / High risk predictions from user-entered road conditions.

eMitra AI – Multi-Source RAG Engine Source Code April 2026
FastAPI, React, Gemini, Qdrant, Crawl4AI

- Built a multi-source Retrieval-Augmented Generation (RAG) engine that answers user queries by combining uploaded documents, crawled website content, and live web search with semantic retrieval from Qdrant.
- Developed an end-to-end ingestion pipeline for website crawling, document parsing, semantic chunking, embedding generation, and vector indexing, enabling low-latency, citation-based responses while reducing LLM hallucinations.

Technical Skills

- **Languages:** C++, JavaScript, Python, SQL
- **Databases:** MongoDB, PostgreSQL, Redis
- **AI / ML:** Machine Learning, Deep Learning, LangChain, LangGraph, Gemini API, Retrieval-Augmented Generation (RAG)
- **Libraries & Frameworks:** React.js, Node.js, REST APIs, Express.js, FastAPI
- **Tools & Cloud:** Git, GitHub, AWS, Docker
- **Core Subjects:** DBMS, OOP, Data Structures & Algorithms, Machine Learning, Operating Systems, Computer Networks

Certifications & Achievements

- **Certification:** Completed **Microsoft** Generative AI Course Certificate
- Solved **800+ DSA** problems across LeetCode and GeeksforGeeks.
- **TechM Code 2026 Finalist** (National-level Hackathon by Tech Mahindra).